1.1 A Prime Factory

WORKSHEET

The question

STEP II 2002, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The last line asks about *infinitely many primes*. The middle asks about *common factors*. What could possibly connect "no two Fermat numbers share a factor" with "there are infinitely many primes"?

STAGE 1 Getting a feel for the numbers

Ask yourself What do you notice about these four numbers?

STAGE 2 Checking the identity

Ask yourself Look at the step from $k = 2$ to $k = 3$. We took 15, multiplied by 17, and got 255. In terms of Fermat numbers, we computed $(F_2 - 2) \times F_2$ and got $F_3 - 2$. Is that a coincidence?

STAGE 3 Proving the identity

Pause and try Before reading on, try to show directly that

$$(F_k - 2)\,F_k = F_{k+1} - 2.$$

Hint: write everything in terms of powers of 2. What does $F_k - 2$ look like?

Ask yourself This is a product of the form $(a-1)(a+1)$. What is that?

Ask yourself What exactly do we need to show in the inductive step?

Ask yourself Could we have avoided induction altogether?

STAGE 4 No common factors

Ask yourself Suppose some number d divides both F_m and F_n, with $m < n$. Where does F_m appear in $(*)$?

Ask yourself Now d divides F_n *and* $F_n - 2$. What else must it divide?

Ask yourself Can d be 2?

STAGE 5 Infinitely many primes

Ask yourself Every whole number greater than 1 has at least one prime factor. Pick one prime factor of each Fermat number. What can you say about the primes you have picked?

What's really going on

Was Fermat right?

No. In 1732 Leonhard Euler showed that

$$F_5 = 2^{32} + 1 = 4\,294\,967\,297 = 641 \times 6\,700\,417.$$

Euler didn't find 641 by luck. He proved that any prime factor of F_n must have a very special form (for $n = 5$, it must be one more than a multiple of 64), which drastically cut down the candidates. You can prove this yourself in Exercise 4.

Since then, every Fermat number anyone has managed to test, from F_5 up to F_{32}, has turned out to be composite. As far as anyone knows, F_0 to F_4 may be the *only* Fermat primes, but nobody has been able to prove it. It is a genuinely open problem.

The delightful twist is that our proof doesn't care. Prime or composite, the Fermat numbers still produce infinitely many primes.

Fermat primes and constructing polygons

Fermat primes turn up somewhere quite unexpected. The ancient Greeks knew how to construct an equilateral triangle and a regular pentagon with ruler and compass, and notice that $3 = F_0$ and $5 = F_1$. For two thousand years nobody could construct any other regular polygon with a prime number of sides. Then in 1796, aged 19, Carl Friedrich Gauss showed how to construct a regular *17-gon*, and $17 = F_2$.

Gauss went further: a regular N-gon can be constructed with ruler and compass exactly when N is a power of 2 multiplied by *distinct Fermat primes*. (Pierre Wantzel completed the proof in 1837.) So which polygons can be constructed depends on how many Fermat primes there are. You will meet the key calculation behind Gauss's 17-gon in Section 6.2 (STEP II 1991, Question 4), and the full story belongs to *Galois theory*, one of the highlights of an undergraduate degree.

One theorem, many proofs

Why would mathematicians bother proving the same theorem more than once? Because different proofs reveal different things. This proof, usually credited to Christian Goldbach in a 1730 letter to Euler, shows that *any* infinite sequence of pairwise coprime whole numbers greater than 1 forces infinitely many primes. Fermat numbers are just one convenient example. (Exercise 3 gives another.)

It also gives more than Euclid's proof does. The primes $p_0, p_1, \ldots, p_{n-1}$ are n *different* primes, each at most F_{n-1}. So there are at least n primes no bigger than $2^{2^{n-1}} + 1$. That is a *quantitative* statement about how the primes are spread out. It is very weak (the truth is much better), but it is a first step towards one of the great results of nineteenth-century mathematics, the *Prime Number Theorem*, which describes exactly how the primes thin out. Section 1.2 takes the next step.

Try this next

1 **Euclid's version.** Write out Euclid's proof that there are infinitely many primes. In what sense is it "by contradiction", and in what sense is the Fermat-number proof "constructive"?

2 **The last digit.** Show that for $n \geq 2$ the Fermat number F_n ends in the digit 7.

3 **Sylvester's sequence.** Define $a_1 = 2$ and $a_{n+1} = a_n^2 - a_n + 1$, so the sequence begins $2, 3, 7, 43, 1807, \ldots$

(a) Show that $a_{n+1} - 1 = a_1 a_2 \cdots a_n$.

(b) Deduce that the terms are pairwise coprime, and hence give another proof that there are infinitely many primes.

(c) HARDER Show that $\dfrac{1}{a_1} + \dfrac{1}{a_2} + \cdots + \dfrac{1}{a_n} = 1 - \dfrac{1}{a_{n+1} - 1}$.

4 **Euler's trick.** *This needs Fermat's Little Theorem: if p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.* Suppose a prime p divides F_n.

(a) Show that $2^{2^n} \equiv -1 \pmod{p}$, and hence that $2^{2^{n+1}} \equiv 1 \pmod{p}$.

(b) Let d be the smallest positive integer with $2^d \equiv 1 \pmod{p}$. Show that d divides 2^{n+1} but not 2^n, and conclude that $d = 2^{n+1}$.

(c) Using Fermat's Little Theorem, show that d divides $p - 1$. Deduce that $p \equiv 1 \pmod{2^{n+1}}$.

(d) For $n = 5$, this says any prime factor of F_5 has the form $64k + 1$. Check that 641 does. How many primes of this form are smaller than 641?

5 **A bound on the primes.** Use the argument in *One theorem, many proofs* to show that the nth prime is at most $2^{2^{n-1}} + 1$. How good is this bound when $n = 5$?

Further reading

- **Martin Aigner & Günter M. Ziegler, *Proofs from THE BOOK*** (Springer), Chapter 1, "Six proofs of the infinity of primes". It includes this proof alongside Euclid's and four others, one of which uses topology. One of the best books you can read before university.
- **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press), Chapters 1–2. A beautifully written introduction to primes and factorisation, ideal for the summer before university.
- **Michal Křížek, Florian Luca & Lawrence Somer, *17 Lectures on Fermat Numbers*** (Springer, 2001). Everything about Fermat numbers, from Euler's factorisation to Gauss's polygons. Dip into the early chapters.
- **Ian Stewart, *Why Beauty Is Truth*** (Basic Books), the chapter on Gauss. The story of the 17-gon told for a general reader.

The full guided solution is in Section 1.1:
https://beyondstep.keithmcnulty.org/ch01/fermat-numbers.html